

Hopf bifurcation problem:

Assume 2-d center manifold at  $\lambda=0$   
with linear problem, after suitable  
linear-transformation

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} p(u,v) \\ q(u,v) \end{pmatrix}$$

$\uparrow$   
 $C^k, k \geq 3$   
 $p, q$  are  $O(2)$   
in  $(u,v)$

Normal Form transformation:

$$\dot{X} = \underbrace{AX}_{\text{linear}} + \underbrace{g_2(X)}_{\text{quadratic}} + \underbrace{g_3(X)}_{\text{cubic}} + \underbrace{o(X^4)}_{\text{remainder}}$$

$$X = \begin{pmatrix} u \\ v \end{pmatrix}$$

$$AX = L = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \omega \begin{pmatrix} -v \\ u \end{pmatrix}$$

$$\text{let } X = Y + \underbrace{P_2(Y)}_{O(Y^2)}$$

first in a  
series of  
"near-identity"  
coordinate  
transformations

Choose  $P_2(Y)$  so that  $Y$ -eqn is "simpler"  
than  $X$ -eqn.

After we do this, we let  $Y = Z + \underbrace{P_3(Z)}_{O(Z^3)}$

and choose  $P_3$  so that  $\dot{Z}$  eqn. is simpler than the  $\dot{Y}$  eqn., etc.

$$\frac{d}{dt} [X = Y + P_2(Y)]$$

~~LHS~~

$$\text{LHS: } \dot{X} = AX + g_2(X) + g_3(X) + O(X^4)$$

$$= A(Y + P_2(Y)) + g_2(Y + P_2(Y)) + g_3(Y + P_2(Y)) + O(Y^4)$$

$$\text{RHS: } \dot{Y} + DP_2(Y) \dot{Y} = (I + DP_2(Y)) \dot{Y}$$

$$\dot{Y} = \underbrace{(I + DP_2(Y))^{-1}}_{(I - DP_2(Y) + O(Y^2))} [AY + AP_2(Y) + g_2(Y) + O(Y^3)]$$

$$\dot{Y} = A\cancel{Y} + \underbrace{-DP_2(Y)AY + AP_2(Y) + g_2(Y)}_{\text{new } O(Y^2) \text{ terms}} + O(Y^3)$$

$g_2(Y)$  is given. We have freedom to choose  $P_2(Y)$ , though.

Could we pick  $P_2(Y)$  so that

$$DP_2 AY - AP_2 = g_2(Y) \quad \text{for any } g_2(Y)?$$

If yes, then  $\dot{Y} = AY + \tilde{g}_3(Y) + O(Y^4)$

ie, we would have succeeded in removing quadratic terms.

$$\underbrace{DP_2 \underbrace{AY}_L - \underbrace{AP_2}_{DL}}_{[L, P_2]} \stackrel{?}{=} g_2(Y)$$

$$[L, P_2] = DP_2 L - DLP_2 = \text{"Lie Bracket"} \\ \text{of vector fields} \\ L, P_2$$

Approach based in linear algebra: consider space of all 2<sup>nd</sup> order vector monomials spanned by

$$\underbrace{\begin{pmatrix} y_1^2 \\ 0 \end{pmatrix}}_{=Y_1}, \underbrace{\begin{pmatrix} y_1 y_2 \\ 0 \end{pmatrix}}_{=Y_2}, \underbrace{\begin{pmatrix} y_2^2 \\ 0 \end{pmatrix}}_{=Y_3}, \underbrace{\begin{pmatrix} 0 \\ y_1^2 \end{pmatrix}}_{=Y_4}, \underbrace{\begin{pmatrix} 0 \\ y_1 y_2 \end{pmatrix}}_{=Y_5}, \underbrace{\begin{pmatrix} 0 \\ y_2^2 \end{pmatrix}}_{=Y_6}$$

$$P_2 = \sum_{j=1}^6 a_j Y_j = a_1 Y_1 + a_2 Y_2 + \dots + a_6 Y_6$$

$$g_2 = \sum_{j=1}^6 b_j Y_j = b_1 Y_1 + b_2 Y_2 + \dots + b_6 Y_6$$

Compute  $[L, Y_j] = DY_j L - DLY_j$   $j=1, \dots, 6$   
 to determine  $[L, P_2] = \sum_{j=1}^6 a_j [L, Y_j]$

$$[L, Y_1] = DY_1 L - DLY_1$$

$$Y_1 = \begin{bmatrix} y_1^2 \\ 0 \end{bmatrix} \quad DY_1 = \begin{bmatrix} 2y_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} -\omega y_2 \\ \omega y_1 \end{bmatrix} \quad DL = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2y_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -\omega y_2 \\ \omega y_1 \end{bmatrix} - \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} y_1^2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -2\omega y_1 y_2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \omega y_1^2 \end{bmatrix} = \begin{bmatrix} -2\omega y_1 y_2 \\ -\omega y_1^2 \end{bmatrix} = -2\omega Y_2 - \omega Y_4$$

$$[L, Y_2] = \omega(Y_1 - Y_3 - Y_5)$$

$$[L, Y_3] = \omega(2Y_2 - Y_6)$$

$$[L, Y_4] = \omega(Y_1 - 2Y_5)$$

$$[L, Y_5] = \omega(Y_2 + Y_4 - Y_6)$$

$$[L, Y_6] = \omega(Y_3 + 2Y_5)$$

$$\omega \left[ \begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ -2 & 0 & 2 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 2 \\ 0 & 0 & -1 & 0 & -1 & 0 \end{array} \right] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix}$$

columns from  $[L, Y]$       depends on  $P_2(Y)$       given by  $g_2(Y)$

$$[L, P_2]^{-1} = g_2$$

yes, ~~since~~ we can choose  $P_2$  to satisfy this eqn.

$$\begin{bmatrix} a_1 \\ \vdots \\ a_6 \end{bmatrix} = \frac{1}{\omega} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ \vdots \\ b_6 \end{bmatrix}$$

can check  $\text{Det} \neq 0$  so invertible.

Now let  $Y = Z + P_3(Z)$  and repeat with

$$\dot{Y} = AY + \tilde{g}_3(Y) + O(Y^4)$$

$$\begin{pmatrix} z_1^3 \\ 0 \end{pmatrix}, \begin{pmatrix} z_1^2 z_2 \\ 0 \end{pmatrix}, \dots \text{ etc }$$

are new  
vector-valued  
monomials

Can we choose  $P_3(\bar{z})$  to remove cubic terms?  
 No, Resulting matrix has a 2-d kernel.  
 We cannot remove

$$(z_1^2 + z_2^2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad (z_1^2 + z_2^2) \begin{pmatrix} -z_2 \\ z_1 \end{pmatrix}$$

Can check

$$\left[ L, \begin{pmatrix} (z_1^2 + z_2^2) z_1 \\ (z_1^2 + z_2^2) z_2 \end{pmatrix} \right] = \left[ L, \begin{pmatrix} (z_1^2 + z_2^2)(-z_2) \\ (z_1^2 + z_2^2)(z_1) \end{pmatrix} \right] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

But these are precisely the terms that have rotational symmetry, from our simple example

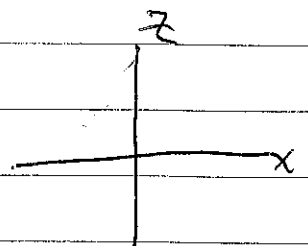
Andrianov-Hopf bifurcation thm. (section 8.8)

Set-up:

$$\begin{aligned} \dot{x} &= -\omega y + p(x, y) \\ \dot{y} &= \omega x + q(x, y) \end{aligned}$$

re-write in complex coordinates

$$\left. \begin{aligned} z &= x + iy \\ \bar{z} &= x - iy \end{aligned} \right\} \Leftrightarrow |z|^2 = x^2 + y^2$$



After normal form transformation & "unfolding"

$$\dot{z} = \lambda(\mu)z + (c(\mu)|z|^2 + d(\mu)|z|^4 + \dots)z$$

$\lambda(0) = i\omega$  for Hopf bifurcation at  $\mu=0$

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta) = x + iy$$

$$\dot{z} = \dot{r}e^{i\theta} + ir\dot{\theta}e^{i\theta} = (\dot{r} + ir\dot{\theta})e^{i\theta}$$

$$= \lambda(\mu)re^{i\theta} + (c(\mu)r^2 + d(\mu)r^4 + \dots)re^{i\theta}$$

$$\dot{r} + ir\dot{\theta} = (\lambda(\mu) + c(\mu)r^2 + d(\mu)r^4 + \dots)r$$

$$\left. \begin{array}{l} \dot{r} = \operatorname{Re} [ \quad ] r \\ \dot{\theta} = \operatorname{Im} [ \quad ] \end{array} \right\} \begin{array}{l} \text{cf. last lecture} \\ c(0) = a + ib \\ \lambda(0) = i\omega \end{array}$$

Hopf bifurcation Thm (8, 21 in text)

Let  $V(x; \mu)$  be  $C^3$  vector field in  $\mathbb{R}^n$  ( $n \geq 2$ ) s.t.

$$V(0; 0) = 0$$

$$\operatorname{spec}(D_x V(0; 0)) = \{i\omega, -i\omega, \lambda_3, \dots, \lambda_n\}, \operatorname{Re}(\lambda_k) < 0, k=3, 4, \dots, n$$

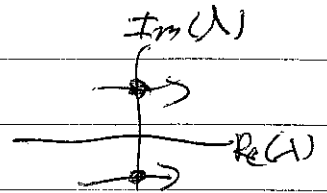
The normal form on the center manifold of  $f_0(\mu=0)$  has an unfolding of the form

$$\dot{z} = \lambda(\mu)z + (c(\mu)|z|^2)z + \dots$$

Assume  $\alpha(0) = \operatorname{Re}(c(0)) \neq 0$

and that a parameter  $\mu$  causes the eigenvalues to cross the imaginary axis i.e.

$$\frac{d}{d\mu} [\operatorname{Re}(\lambda(\mu))]_{\mu=0} \neq 0$$



Then there is a ~~Hopf~~ bifurcation that gives birth to a limit cycle in the center manifold at  $\mu=0$ ,

super vs. sub Hopf { The limit cycle exists when  $\alpha(\lambda) < 0$  and is stable in center manifold if  $\operatorname{Re}(\lambda) > 0$  & unstable if  $\operatorname{Re}(\lambda) < 0$ ,

~~super~~

Note: There is an explicit formula for  $\alpha(0)$  in terms of quadratic & cubic coeffs. of  $p$  &  $q$  in text:

$$\begin{aligned}\dot{x} &= -\omega y + p(x, y) \\ \dot{y} &= \omega x + q(x, y)\end{aligned}$$